

SQL MASTERY — REMAINING ROADMAP

From your current level → Advanced SQL → Interview Mastery

CURRENTLY COMPLETED

- ✓ SQL fundamentals and MySQL Workbench
- ✓ CREATE DATABASE, CREATE TABLE, INSERT
- ✓ SELECT and selecting specific columns
- ✓ WHERE and comparison operators
- ✓ AND, OR, NOT
- ✓ IN, NOT IN
- ✓ BETWEEN, NOT BETWEEN
- ✓ LIKE with % and _
- ✓ DISTINCT
- ✓ ORDER BY
- ✓ LIMIT and OFFSET

REMAINING ROADMAP

LEVEL 2 — Aggregate SQL

- COUNT()
- SUM()
- AVG()
- MIN()
- MAX()
- Aggregate functions + WHERE
- GROUP BY
- Multiple-column GROUP BY
- HAVING
- WHERE vs HAVING
- Aggregate functions + ORDER BY

LEVEL 3 — Database Relationships & Keys

- Primary Key
- Foreign Key
- Candidate Key
- Composite Key
- One-to-One
- One-to-Many
- Many-to-Many

LEVEL 4 — JOINS ■■■

- Why JOINS are needed
- INNER JOIN
- LEFT JOIN
- RIGHT JOIN
- FULL OUTER JOIN concept
- SELF JOIN
- CROSS JOIN
- Joining 3+ tables
- JOIN + WHERE
- JOIN + GROUP BY
- JOIN + HAVING
- JOIN + ORDER BY

LEVEL 5 — Subqueries ■■■

- What is a subquery?
- Single-row subquery
- Multi-row subquery
- Subquery with IN
- EXISTS
- NOT EXISTS
- Subquery with FROM
- Subquery with SELECT
- Correlated subquery
- Nested subqueries

LEVEL 6 — Conditional SQL

- CASE
- Simple CASE
- Searched CASE
- CASE with aggregation
- Conditional counting
- Conditional SUM
- COALESCE
- NULLIF

LEVEL 7 — Set Operations

- UNION
- UNION ALL

- INTERSECT concept
- EXCEPT concept
- Set-operation rules
- Set operations with filtering

LEVEL 8 — NULL Handling

- What is NULL?
- IS NULL
- IS NOT NULL
- NULL in calculations
- NULL with aggregate functions
- COALESCE
- NULL in JOINS
- Common NULL mistakes

LEVEL 9 — Constraints

- PRIMARY KEY
- FOREIGN KEY
- NOT NULL
- UNIQUE
- DEFAULT
- CHECK
- Referential integrity
- ON DELETE
- ON UPDATE
- Cascading

LEVEL 10 — Database Design

- ER diagrams
- Entities and attributes
- Relationships
- Cardinality
- Normalization
- 1NF
- 2NF
- 3NF
- BCNF
- Denormalization

LEVEL 11 — Advanced SQL

- Views
- Creating and using views
- Stored Procedures
- Parameters
- Stored Functions
- Triggers
- Before/After triggers
- Cursors

LEVEL 12 — Transactions

- What is a transaction?
- COMMIT
- ROLLBACK
- SAVEPOINT
- ACID properties
- Atomicity
- Consistency
- Isolation
- Durability
- Transaction problems

LEVEL 13 — Window Functions ■■■

- Window functions
- OVER()
- PARTITION BY
- ROW_NUMBER()
- RANK()
- DENSE_RANK()
- LEAD()
- LAG()
- Running totals
- Running averages
- Moving averages
- Ranking within departments
- Top-N per group

LEVEL 14 — CTEs ■■■

- Common Table Expressions
- WITH

- Multiple CTEs
- CTE + JOIN
- CTE + aggregation
- CTE + window functions
- Recursive CTE

LEVEL 15 — SQL Performance

- Indexes
- Why indexes matter
- Creating indexes
- Single-column index
- Composite index
- Index advantages/disadvantages
- EXPLAIN
- Execution plans
- Full table scans
- Query optimization
- Common performance mistakes

LEVEL 16 — Interview Problem Solving ■

- Second-highest salary
- Nth-highest salary
- Duplicate records
- Delete duplicates
- Employees above average salary
- Employees above department average
- Highest salary per department
- Top 3 salaries per department
- Employees with no manager
- Employees earning more than manager
- Customers who never ordered
- Products never ordered
- Most frequently ordered product
- Monthly sales
- Monthly revenue
- Running revenue
- Consecutive dates
- Missing dates
- Duplicate transactions

- Customer retention
- Ranking problems
- Gaps and islands
- Employee hierarchy
- Complex multi-table problems

45-DAY RECOMMENDED PLAN

Period	Topics
Days 1–7	Aggregate functions, GROUP BY, HAVING, NULL, CASE
Days 8–17	Keys, relationships, JOINS, multi-table problems
Days 18–24	Subqueries, correlated subqueries, EXISTS, NOT EXISTS
Days 25–31	CTEs and window functions
Days 32–35	Views, procedures, functions, triggers, transactions
Days 36–38	Indexes, EXPLAIN, optimization, database design
Days 39–45	Interview problems, mixed SQL tests, mock interviews

MASTER SQL TARGET

Target: 45 days with approximately 1–1.5 hours of focused practice per day. The goal is not to memorize syntax. For every topic, learn the concept, practice it in MySQL Workbench, solve progressively harder problems, and revisit mistakes.

Final outcome: strong placement-level SQL plus advanced skills in JOINS, subqueries, window functions, CTEs, optimization, and real-world SQL problem solving.